

THREE

A MATHEMATICAL
MEDITATION ON HISTORY

On Monte Carlo simulation as a metaphor to understanding a sequence of random historical events. On randomness and artificial history. Age is beauty, almost always, and the new and the young are generally toxic. Send your history professor to an introductory class on sampling theory.

EUROPLAYBOY MATHEMATICS

The stereotype of a pure mathematician presents an anemic man with a shaggy beard and grimy and uncut fingernails silently laboring on a Spartan but disorganized desk. With thin shoulders and a pot belly, he sits in a grubby office, totally absorbed in his work, oblivious to the grunginess of his surroundings. He grew up in a communist regime and speaks English with an astringent and throaty Eastern European accent. When he eats, crumbs of food accumulate in his beard. With time he becomes more and more absorbed in his subject matter of pure theorems, reaching levels of ever increasing abstraction. The American public was recently exposed to one of these characters with the *unabomber*, the bearded and recluse mathematician who lived in a hut and took to murdering people who promoted modern

technology. No journalist was capable of even coming close to describing the subject matter of his thesis, *Complex Boundaries*, as it has no intelligible equivalent – a complex number being an entirely abstract and imaginary number, the square root of minus one, an object that has no analog outside of the world of mathematics.

The name Monte Carlo conjures up the image of a suntanned urbane man of the Europlayboy variety entering a casino under a whiff of the Mediterranean breeze. He is an apt skier and tennis player, but also can hold his own in chess and bridge. He drives a gray sports car, dresses in a well ironed Italian handmade suit, and speaks carefully and smoothly about mundane, but real, matters, those a journalist can easily describe to the public in compact sentences. Inside the casino he astutely counts the cards, mastering the odds, and bets in a studied manner, his mind producing precise calculations of his optimal betting size. He could be James Bond's smarter lost brother.

Now when I think of Monte Carlo mathematics, I think of a happy combination of the two: the Monte Carlo man's realism without the shallowness combined with the mathematician's intuitions without the excessive abstraction. For indeed this branch of mathematics is of immense practical use – it does not present the same dryness commonly associated with mathematics. I became addicted to it the minute I became a trader. It shaped my thinking in most matters related to randomness. Most of the examples used in the book were created with my Monte Carlo generator, which I introduce in this chapter. Yet, it is far more a way of thinking than a computational method. Mathematics is principally a tool to meditate, rather than to compute.

THE TOOLS

The notion of alternative histories discussed in the last chapter can be extended considerably and subjected to all manner of technical refinement. This brings us to the tools used in my profession to toy with uncertainty. I will outline them next. Monte Carlo methods, in brief, consist in creating artificial history using the following concepts.

First, consider the sample path. The invisible histories have a scientific name, *alternative sample paths*, a name borrowed from the field of